

Database Manager : Volume Reactions

FileEditDatabaseUnits

Volume Reactions

reactions_vol_2

Local

AlN_volume_BO[CH

BN_Allendorf

BN_Allendorf_SIMA

Chlorine_radicals

CrC_from_bisbenze

SiH4+ C3H8(Dron)

ACE

CVD

PLASMA

Name

reactions_vol_2

Notes

Mechanism Type: Finite-Rate (Species Approach in Gas Phase) - Multi

Action Type: General Rates

Units: SI

Reaction Steps

No	Equation	Notes	Apf	nf	mf	(Ea/R)f	Apb	nb	mb	(Ea/R)b
1	CO(ACAC)3+M->CO(ACAC)2+M+ACAC		2460000000000	0.5	0	13455				
2	ZO->O2		1030000000000	0.5	0	0				

Add

Copy

Paste

Remove

Reaction Step Properties

Step 1

Equation

CO(ACAC)3+M->CO(ACAC)2+M+ACAC

Name

Notes

Define Third Body Efficiencies

Pressure Dependent

☐

Step Type

Arrhenius

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap)

2460000000000

Temperature Exponent (n)

0.5

Pressure Exponent (m)

0

Activation Temperature (Ea/R)

13455

User Sub (ureaction_rate_factor)

☐

Forward Concentration Exponents

1

1

Apply

Reset

Ready.

Rechercher

15:17

01/12/2025

Database Manager : Surface Reactions

File Edit Database Units

Surface Reactions

Local

- tubev2
 - reaction_surf2
- Local
 - AIN_surface_BOICH
 - BN_surface_reaction
 - BN_surface_reaction
 - CrC_from_bisbenze
 - NC-face(Nishizawa)
 - TiN_growth_by_Jua
- ACE
 - PLASMA
 - CVD

Name: reaction_surf2

Notes: On calcule les AP à l'aide de la formule des réactions unimoléculaire puis o

Mechanism Type: Finite-Rate (Species Approach in Gas Phase) - Multi Units: SI

Surface Site

- tubev2
 - surface_react
- Local
- BN surface sites

Add Remove

surface_react

Bulk Species

- COO(B)
- ALN(B)
- C(B)
- GAAS(B)

Surface Species

- CO(S)
- O(S)

Species

- CO(ACAC)3
- CO(ACAC)2
- O
- O2

Reaction Steps

No	Equation	Notes	Type	Apf	nf	(Ea/R)f	Apb	nb	(Ea/R)b
1	CO(ACAC)2+O(S)->COO(B)+CO(S)+2A...		(FR)	2474000000	0.5	0			
2	O2+CO(S)->O(S)+O		(FR)	6400000000	0.5	0			

Add Copy Paste Remove

Reaction Step Properties

Step 1

Equation: CO(ACAC)2+O(S)->COO(B)+CO(S)+2ACAC

Name

Notes

Surface Coverage Modification ☐

Electro Chemical No

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap) 2474000000

Temperature Exponent (n) 0.5

Activation Temperature (Ea/R) 0

User Sub (ureaction_rate_factor) ☐

Sticking Coefficient No

Ion Energy Dependent

Ion Energy Dependent Rate Factor ☐

Ion Energy Dependent Yield Factor ☐

Ion Angle Dependent Yield Factor ☐

Forward Concentration Exponents

1

1

Apply Reset

Ready.

Rechercher

15:17 01/12/2025

Database Manager: Surface Reactions

File Edit Database Units

Surface Reactions

Tube_simple_AIN_growth

AIN_surface_BOICHOT

Local

AIN_surface_BOICHOT

BN_surface_reaction

BN_surface_reaction_SI

CrC_from_bisbenzene

NC-face(Nishizawa)

TiN_growth_by_Juan_S

ACE

PLASMA

CVD

Name: AIN_surface_BOICHOT_JCMCTF

Notes: <https://doi.org/10.1007/s00214-013-1419-8>

Mechanism Type: Finite-Rate (Species Approach in Gas Phase) - Multi Units: SI

Surface Site

Tube_simple_AIN_growth

AIN_surface

Local

BN surface sites

Bulk Species

ALN(B)

C(B)

GAAS(B)

GAN(B)

Surface Species

N(S)

AL(S)

Species

AC7H14

ADDUCT

AL

ALCL3

Reaction Steps

No	Equation	Notes	Type	Apf	nf	(Ea/R)f	Apb	nb	(Ea/R)b
1	NH3+AL(S)<->N(S)+H2+H		(FR)	668000000	0.5	8654	0	0	0
2	ALCL3+N(S)<->ALN(B)+AL(S)+CL+CL2		(FR)	238500000	0.5	7700	0	0	0
3	ALCL2NH2+AL(S)<->ALN(B)+AL(S)+2H...		(FR)	258000000	0.5	5100	0	0	0
4	ALCL2NH2+N(S)<->ALN(B)+N(S)+2HCL		(FR)	258000000	0.5	5100	0	0	0

Reaction Step Properties

Step 1

Equation: NH3+AL(S)<->N(S)+H2+H

Name:

Notes:

Surface Coverage Modification: ☐

Electro Chemical: No

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap): 668000000

Temperature Exponent (n): 0.5

Activation Temperature (Ea/R): 8654

User Sub (ureaction_rate_factor): ☐

Sticking Coefficient: No

Backward Reaction Rate Constants

Pre-Exponential Factor (Ap): 0

Temperature Exponent (n): 0

Activation Temperature (Ea/R): 0

User Sub (ureaction_rate_factor): ☐

Ion Energy Dependent

Ion Energy Dependent Rate Factor: ☐

Ion Energy Dependent Yield Factor: ☐

Ion Angle Dependent Yield Factor: ☐

Forward Concentration Exponents

1

1

Apply Reset

Ready.

Rechercher

14:39 01/12/2025

Database Manager : Volume Reactions

File Edit Database Units

Volume Reactions

Tube_simple_AIN_growth

AIN_volume_BOICHOT

Local

AIN_volume_BOICHOT

BN_Allendorf

BN_Allendorf_SIMAP_f

Chlorine_radicals

CrC_from_bisbenzene_

SiH4+C3H8(Dron)

ACE

CVD

PLASMA

Name: AIN_volume_BOICHOT_JCMCTF

Notes: <https://doi.org/10.1007/s00214-013-1419-8>

Mechanism Type: Finite-Rate (Species Approach in Gas Phase) - Multi

Action Type: General Rates

Units: CGS

Reaction Steps

No	Equation	Notes	Apf	nf	mf	(Ea/R)f	Apb	nb	mb	(Ea/R)b
1	$\text{ALCL}_3 + \text{NH}_3 \leftrightarrow \text{ALCL}_2\text{NH}_2 + \text{HCL}$	hypothe...	421000000000	0	0	4198				
2	$\text{H} + \text{CL} + \text{M} \rightarrow \text{HCL} + \text{M}$	Allendorf	7.2E+21	-2	0	0	7.9E+25	-3	0	53598
3	$2\text{CL} + \text{M} \rightarrow \text{CL}_2 + \text{M}$	Allendorf	2E+14	0	0	-901				
4	$\text{H} + \text{HCL} \rightarrow \text{CL} + \text{H}_2$	Allendorf	1.69E+13	0	0	2082	2.95E+13	0	0	2567
5	$\text{CL}_2 + \text{H} \rightarrow \text{CL} + \text{HCL}$	Allendorf	8.6E+13	0	0	590				
6	$2\text{H} + \text{H}_2 \rightarrow 2\text{H}_2$	Allendorf	9.7E+16	-0.6	0	0	8.8E+14	0	0	48364
7	$\text{H}_2 + \text{M} \rightarrow 2\text{H} + \text{M}$	Allendorf	2.2E+14	0	0	48364	6.53E+17	-1	0	0

Add Copy Paste Remove

Reaction Step Properties

Step 1

Equation: $\text{ALCL}_3 + \text{NH}_3 \leftrightarrow \text{ALCL}_2\text{NH}_2 + \text{HCL}$

Name:

Notes: hypothesis AI

Step Type: Arrhenius

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap): 421000000000

Temperature Exponent (n): 0

Pressure Exponent (m): 0

Activation Temperature (Ea/R): 4198

User Sub (ureaction_rate_factor): ☐

Backward Reaction Rate Constants

Backward Rate: By Equilibrium

Forward Concentration Exponents

1

1

Backward Concentration Exponents

1

1

Apply Reset

Ready.

Rechercher

14:47 01/12/2025

Database Manager : Surface Reactions

File Edit Database Units

Surface Reactions

Tube_simple_AIN_growth
AIN_surface_BOICHOT
Tube_simple_TiN_growth
TiN_growth_by_Juan_S

Local
AIN_surface_BOICHOT
BN_surface_reaction
BN_surface_reaction_SI
CrC_from_bisbenzene
NC-face(Nishizawa)
TiN_growth_by_Juan_S

ACE
PLASMA
CVD

Name: TiN_growth_by_Juan_S_SIMAP
Notes: <https://doi.org/10.1039/C9CE00488B>

Mechanism Type: Finite-Rate (Species Approach in Gas Phase) - Multi Units: CGS

Surface Site

Tube_simple_TiN_growth
TiN

Add Remove

Bulk Species

ALN(B)
C(B)
GAAS(B)
GAN(B)

Surface Species

N(S)
Ti(S)

Species

AC7H14
ADDUCT
AL
ALCL3

Reaction Steps

No	Equation	Notes	Type	Apf	nf	(Ea/R)f	Apb	nb	(Ea/R)b
1	TiCl3+N(S)->Ti(S)+CL2+CL	N deposi... (FR)		1E+14	0	4900			
2	NH3+Ti(S)->TiN(B)+N(S)+H2+H	Ti depos... (FR)		240000000000	0	4900			

Add Copy Paste Remove

Reaction Step Properties

Step 1

Equation: TiCl3+N(S)->Ti(S)+CL2+CL

Name:

Notes: N deposition

Surface Coverage Modification: ☒

Specify Profile...

Electro Chemical: No

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap): 1E+14

Temperature Exponent (n): 0

Activation Temperature (Ea/R): 4900

User Sub (ureaction_rate_factor): ☐

Sticking Coefficient: No

Ion Energy Dependent

Ion Energy Dependent Rate Factor: ☐

Ion Energy Dependent Yield Factor: ☐

Ion Angle Dependent Yield Factor: ☐

Forward Concentration Exponents

1

1

Apply Reset

Ready.

Rechercher

14:49
01/12/2025

Database Manager : Volume Reactions

File Edit Database Units

Volume Reactions

Tube_simple_AIN_growth

AIN_volume_BOICHOT

Tube_simple_TiN_growth

Chlorine_radicals

Local

AIN_volume_BOICHOT

BN_Allendorf

BN_Allendorf_SIMAP_f

Chlorine_radicals

CrC_from_bisbenzene_

SiH4+ C3H8(Dron)

ACE

CVD

PLASMA

NameChlorine_radicals

Noteshttps://doi.org/10.1007/s00214-013-1419-8

Mechanism Type:Finite-Rate (Species Approach in Gas Phase) - Multi

Action Type:General Rates

Units:CGS

Reaction Steps

No	Equation	Notes	Apf	nf	mf	(Ea/R)f	Apb	nb	mb	(Ea/R)b
1	$H + CL + M \rightleftharpoons HCL + M$	Allendorf	7.2E+21	-2	0	0	7.9E+25	-3	0	53598
2	$2CL + M \rightleftharpoons CL_2 + M$	Allendorf	2E+14	0	0	-901				
3	$H + HCL \rightleftharpoons CL + H_2$	Allendorf	1.69E+13	0	0	2082	2.95E+13	0	0	2567
4	$CL_2 + H \rightleftharpoons CL + HCL$	Allendorf	8.6E+13	0	0	590				
5	$2H + H_2 \rightleftharpoons 2H_2$	Allendorf	9.7E+16	-0.6	0	0	8.8E+14	0	0	48364
6	$H_2 + M \rightleftharpoons 2H + M$	Allendorf	2.2E+14	0	0	48364	6.53E+17	-1	0	0

Add

Copy

Paste

Remove

Reaction Step Properties

Step 1

Equation

H+CL+M<->HCL+M

Name

Notes

Allendorf

Define Third Body Efficiencies

Pressure Dependent

☐

Step Type

Arrhenius

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap)

7.2E+21

Temperature Exponent (n)

-2

Pressure Exponent (m)

0

Activation Temperature (Ea/R)

0

User Sub (ureaction_rate_factor)

☐

Backward Reaction Rate Constants

Backward Rate

Specified

Pre-Exponential Factor (Ap)

7.9E+25

Temperature Exponent (n)

-3

Pressure Exponent (m)

0

Activation Temperature (Ea/R)

53598

User Sub (ureaction_rate_factor)

☐

Forward Concentration Exponents

1

1

1

Backward Concentration Exponents

1

Apply

Reset

Ready.

Rechercher

14:5001/12/2025

Database Manager : Surface Reactions

File Edit Database Units

Surface Reactions

Tube_simple_AIN_growth
AIN_surface_BOICHOT_ICMCTF
Tube_simple_TiN_growth
TiN_growth_by_Juan_Su_SIMAP
Tube_simple_SiC_growth
NC-face(Nishizawa)
Tube_simple_CrC_growth
CrC_from_bisbenzene_Chromium_
Local
AIN_surface_BOICHOT_ICMCTF
BN_surface_reaction
BN_surface_reaction_SIMAP_fit_OK
CrC_from_bisbenzene_Chromium_
NC-face(Nishizawa)
TiN_growth_by_Juan_Su_SIMAP

ACE
PLASMA
CVD

Name: NC-face(Nishizawa)
Notes:

Mechanism Type: Finite-Rate (Species Approach in Gas Phase) - Multi Units: CGS

Surface Site
Tube_simple_SiC_growth
SICN
Local
BN surface sites

Bulk Species
ALN(B)
C(B)
GAAS(B)
GAN(B)

Surface Species
Si_S(S)
SiH2_S(S)
C_S(S)
N_S(S)

Species
AC7H14
ADDUCT
AL
ALCL3

Reaction Steps

No	Equation	Notes	Type	Apf	nf	(Ea/R)f	Apb	nb	(Ea/R)b
1	$C(B) + Si_S(S) + H_2 \rightarrow SiH_2 + C_S(S)$		(FR)	2.2E+32	0	52786			
2	$2Si(B) + 2C_S(S) + H_2 \rightarrow C_2H_2 + 2Si_S(S)$		(FR)	2.2E+26	0	52786			
3	$SiH_4 + C_S(S) \rightarrow SiH_2_S(S) + H_2 + C(B)$		(FR)	15920000000	0	9401			
4	$SiH_2_S(S) \rightarrow H_2 + Si_S(S)$	SiH2_S -> H2 + Si_S	(FR)	2.912E+14	0	4527			
5	$SiH_2 + C_S(S) \rightarrow SiH_2_S(S) + C(B)$	SiH2 + C_S -> SiH2_S + C	(FR)	3060000000000	0.5	0			
6	$Si_G + C_S(S) \rightarrow Si_S(S) + C(B)$	Si_G + C_S -> Si_S + C	(FR)	3167000000000	0.5	0			
7	$C_2H_2 + 2Si_S(S) \rightarrow 2C_S(S) + H_2 + 2Si(B)$	C2H2 + 2Si_S -> 2C_S + H2 + 2Si	(FR)	3.04E+40	0.5	0			
8	$C_2H_4 + 2Si_S(S) \rightarrow 2C_S(S) + 2H_2 + 2Si(B)$	C2H4 + 2Si_S -> 2C_S + 2H2 + 2Si	(FR)	2.342E+38	0.5	0			
9	$CH_4 + Si_S(S) \rightarrow C_S(S) + 2H_2 + Si(B)$	CH4 + Si_S -> C_S + 2H2 + Si	(FR)	209850	0.5	0			
10	$HSiCH_3 + C_S(S) \rightarrow C(B) + Si_S(S) + H + CH_3$		(FR)	2490000000000	0.5	0			
11	$CH_3 + Si_S(S) \rightarrow Si(B) + C_S(S) + 1.5H_2$		(FR)	4270000000000	0.5	0			
12	$Si_2 + 2C_S(S) \rightarrow 2C(B) + 2Si_S(S)$		(FR)	1000000000000	0.5	0			
13	$Si_2C + Si_S(S) \rightarrow Si_2 + Si(B) + C_S(S)$		(FR)	1E+25	0.5	0			
14	$SiCH_2 + C_S(S) \rightarrow C(B) + Si_S(S) + CH_2$		(FR)	2550000000000	0.5	0			
15	$CH_2 + Si_S(S) \rightarrow C(B) + Si_S(S) + H_2$		(FR)	4419000000000	0.5	0			
16	$N_2 + 2Si_S(S) \rightarrow 2N_S(S) + 2Si(B)$		(FR)	1E+15	0	-31500			
17	$N_S(S) \rightarrow N_V(B)$		(SC)	1	0	0			

Add Copy Paste Remove

Reaction Step Properties

Step 1

Equation: $C(B) + Si_S(S) + H_2 \rightarrow SiH_2 + C_S(S)$

Name:

Notes:

Surface Coverage Modification: ☐

Electro Chemical: No

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap): 2.2E+32

Temperature Exponent (n): 0

Activation Temperature (Ea/R): 52786

User Sub (ureaction_rate_factor): ☐

Sticking Coefficient: No

Ion Energy Dependent

Ion Energy Dependent Rate Factor: ☐

Apply Reset

Ready.

Rechercher

14:56
01/12/2025

Database Manager : Volume Reactions

File Edit Database Units

Volume Reactions

Tube_simple_AIN_growth
AIN_volume_BOICHOT_ICMCTF
Tube_simple_TiN_growth
Chlorine_radicals
Tube_simple_SiC_growth
SiH4+ C3H8(Dron)
Tube_simple_CrC_growth
CrC_from_bisbenzene_Chromium_

Local
AIN_volume_BOICHOT_ICMCTF
BN_Allendorf
BN_Allendorf_SIMAP_fit_OK
Chlorine_radicals
CrC_from_bisbenzene_Chromium_
SiH4+ C3H8(Dron)

ACE
CVD
PLASMA

Name: SiH4+ C3H8(Dron)
Notes: Chemical

Mechanism Type: Finite-Rate (Species Approach in Gas Phase) - Multi Action Type: General Rates Units: CGS

Reaction Steps

No	Equation	Notes	Apf	nf	mf	(Ea/R)f	Apb	nb	mb	(Ea/R)b
1	SiH4 <-> SiH2 + H2	SiH4 <-> SiH2 + H2	6.671E+29	-4.795	0	31946				
2	Si2H6 <-> SiH2 + SiH4	Si2H6 <-> SiH2 + SiH4	3.24E+29	-4.24	0	29202				
3	SiH2 <-> Si_G + H2	SiH2 <-> Si_G + H2	1.06E+14	-0.88	0	22657				
4	2H + H2 <-> 2H2	2H + H2 <-> 2H2	9.2E+16	-0.6	0	0				
5	C3H8 <-> CH3 + C2H5	C3H8 <-> CH3 + C2H5	1.698E+16	0	0	42715				
6	CH4 + H <-> CH3 + H2	CH3 + H2 <-> CH4 + H	22000	3	0	4406				
7	C2H5 + H <-> 2CH3	2CH3 <-> C2H5 + H	1E+14	0	0	13350				
8	2CH3 <-> C2H6	2CH3 <-> C2H6	9.03E+16	-1.18	0	330				
9	C2H4 + H <-> C2H5	C2H5 <-> C2H4 + H	2.21E+13	0	0	1040				
10	C2H4 <-> C2H2 + H2	C2H4 <-> C2H2 + H2	1.5E+15	0	0	41467				
11	H3SiCH3 <-> SiH2 + CH4		2E+14	0	0	36413				
12	H3SiCH3 <-> HSiCH3 + H2		1E+14	0	0	32033				
13	Si2 <-> 2Si_G		1E+15	0	0	37460				
14	Si2 + CH4 <-> Si2C + 2H2		3.011E+15	0	0	10000				
15	SiH2 + Si_G <-> Si2 + H2		1.5E+14	0	0	0				
16	CH3 + Si_G <-> SiCH2 + H		1.39E+12	0.5	0	0				
17	SiCH2 + SiH2 <-> Si2C + 2H2		1E+15	0	0	0				

Add Copy Paste Remove

Reaction Step Properties

Step 1

Equation: SiH4 <-> SiH2 + H2
Name: SiH4 <-> SiH2 + H2
Notes: SiH4 <-> SiH2 + H2
Step Type: Arrhenius

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap): 6.671E+29
Temperature Exponent (n): -4.795
Pressure Exponent (m): 0
Activation Temperature (Ea/R): 31946
User Sub (ureaction_rate_factor): ☐

Backward Reaction Rate Constants

Backward Rate: By Equilibrium

Forward Concentration Exponents: 1
Backward Concentration Exponents: 1

Apply Reset

Ready.

Rechercher

14:57
01/12/2025

Database Manager : Surface Reactions

File Edit Database Units

Surface Reactions

Name: CrC_from_bisbenzene_Chromium_surface
Notes: <https://doi.org/10.3390/coatings8060220>

Mechanism Type: Finite-Rate (Species Approach in Gas Phase) - Multi Units: SI

Surface Site

Tube_simple_CrC_growth
CR_SITE
Local
BN surface sites

Bulk Species

ALN(B)
C(B)
GAAS(B)
GAN(B)

Surface Species

M_CR(S)
M_C6H6(S)
M_C6H5SH(S)
M_C6CL6(S)

Species

AC7H14
ADDUCT
AL
ALCL3

Reaction Steps

No	Equation	Notes	Type	Apf	nf	(Ea/R)f	Apb	nb	(Ea/R)b
1	MH_MBC+M_CR(S)->M_CR(B)+M_CR(S)+MH_C6H6		(FR)	2.5E+15	0.5	8000			
2	MH_C6H6+M_CR(S)->2.3C(B)+1.15M_H...		(SC)	1	0	0			

Reaction Step Properties

Step 1

Equation: MH_MBC+M_CR(S)->M_CR(B)+M_CR(S)+MH_C6H6

Name:

Notes:

Surface Coverage Modification: ☐

Electro Chemical: No

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap): 2.5E+15

Temperature Exponent (n): 0.5

Activation Temperature (Ea/R): 8000

User Sub (ureaction_rate_factor): ☐

Sticking Coefficient: No

Ion Energy Dependent

Ion Energy Dependent Rate Factor: ☐

Ion Energy Dependent Yield Factor: ☐

Ion Angle Dependent Yield Factor: ☐

Forward Concentration Exponents

1

1

Ready.

Rechercher

14:54 01/12/2025

Database Manager : Volume Reactions

File Edit Database Units

Volume Reactions

Name: CrC_from_bisbenzene_Chromium_volume
Notes: <https://doi.org/10.3390/coatings8060220>
Modified: February 19, 2024

Mechanism Type: Finite-Rate (Species Approach in Gas Phase) - Multi Action Type: General Rates Units: SI

Reaction Steps

No	Equation	Notes	Apf	nf	mf	(Ea/R)f	Apb	nb	mb	(Ea/R)b
1	MH_BBC->MH_MBC+MH_C6H6_BBC		80000	0.5	0	7000				

Add Copy Paste Remove

Reaction Step Properties

Step 1

Equation: MH_BBC->MH_MBC+MH_C6H6_BBC

Name:

Notes:

Step Type: Arrhenius

Forward Reaction Rate Constants

Pre-Exponential Factor (Ap): 80000

Temperature Exponent (n): 0.5

Pressure Exponent (m): 0

Activation Temperature (Ea/R): 7000

User Sub (ureaction_rate_factor): ☐

Forward Concentration Exponents

1

Apply Reset

Ready.

Rechercher

14:58 01/12/2025

Database Manager : Species

File Edit Database Units

Species

Name: MH_BBC

Notes

Elemental Composition Properties

Molecular
Physical
Fluid
Thermal
Chemical

Thermal Accomodation Coefficient
Thermal Accomodation Coefficient 0.9317

Specific Heat
JANNAF Coefficients
$$\frac{C_p}{R} = a_1 + a_2 T + a_3 T^2 + a_4 T^3 + a_5 T^4$$

$$\frac{H}{RT} = a_1 + \frac{a_2 T}{2} + \frac{a_3 T^2}{3} + \frac{a_4 T^3}{4} + \frac{a_5 T^4}{5} + \frac{a_6}{T}$$

$$\frac{S}{R} = a_1 \ln T + a_2 T + \frac{a_3 T^2}{2} + \frac{a_4 T^3}{3} + \frac{a_5 T^4}{4} + a_7$$

Temperature Limits
Lower Limit 298 K
Break Point 660 K
Upper Limit 1000 K
Coefficients at Lower Limit
a1 -12.065
a2 0.1459584
a3 -0.0001248
a4 6.8427E-08
a5 -2.9991E-11
a6 27931.2004
a7 387.51
Coefficients at Upper Limit
a1 7.133
a2 0.07445
a3 -2.73E-05
a4 1.095E-17
a5 -6.776E-21
a6 23990
a7 294.2
Polynomial Coefficients in Temperature
The following Polynomial is used when Mix Polynomial in T is selected as evaluation method.
$$f(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4 + c_5 T^5$$

C0 0
C1 0
C2 0
C3 0
C4 0
C5 0
Mix Linear in T
Mix Linear in T
Conductivity
Polynomial Coefficients in Temperature
The following Polynomial is used when

Ready.

15:01
01/12/2025

Database Manager : Species

File Edit Database Units

Species

Name: MH_MBC

Notes

Elemental Composition Properties

Molecular
Physical
Fluid
Thermal
Chemical

Thermal Accomodation Coefficient

Thermal Accomodation Coefficient 0.9317

Specific Heat

JANNAF Coefficients

$$\frac{C_p}{R} = a_1 + a_2 T + a_3 T^2 + a_4 T^3 + a_5 T^4$$
 Plot

$$\frac{H}{RT} = a_1 + \frac{a_2 T}{2} + \frac{a_3 T^2}{3} + \frac{a_4 T^3}{4} + \frac{a_5 T^4}{5} + \frac{a_6}{T}$$
 Plot

$$\frac{S}{R} = a_1 \ln T + a_2 T + \frac{a_3 T^2}{2} + \frac{a_4 T^3}{3} + \frac{a_5 T^4}{4} + a_7$$
 Plot

Temperature Limits

Lower Limit 298 K

Break Point 770 K

Upper Limit 1000 K

Coefficients at Lower Limit

a1 -8.2912

a2 0.09563

a3 -0.00011765

a4 9.4277E-08

a5 -3.7156E-11

a6 29927.8326

a7 315.23

Coefficients at Upper Limit

a1 2.995

a2 0.04138

a3 -1.576E-05

a4 -2.168E-19

a5 0

a6 28090

a7 49.3469182

Polynomial Coefficients in Temperature

The following Polynomial is used when Mix Polynomial in T is selected as evaluation method.

$$f(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4 + c_5 T^5$$

C0 0

C1 0

C2 0

C3 0

C4 0

C5 0

Mix Linear in T

Mix Linear in T

Conductivity

Polynomial Coefficients in Temperature

The following Polynomial is used when

Ready.

Rechercher

15:02
01/12/2025